Fiscal Year:	FY 2021	**Task Last Updated:**	FY 08/02/2021
PI Name:	Blancaflor, Elison B. Ph.D.		
Project Title:	The Use of Microgravity Simulators for a Mechanistic Understanding of Cytoskeletal-Mediated Regulation of Root Growth		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
PI Email:	elison.b.blancaflor@nasa.gov	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	321-501-5368
Organization Name:	NASA Kennedy Space Center		
PI Address 1:	Utilization and Life Science Office		
PI Address 2:	Exploration and Research Technology Programs		
PI Web Page:			
City:	Kennedy Space Center	**State:**	FL
Zip Code:	32955	**Congressional District:**	8
Comments:	NOTE: Dr. Blancaflor moved to NASA Kennedy Space Center in spring 2021; as a PI, his prior grants were for work at the Noble Foundation in Oklahoma.		
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
Start Date:	07/26/2018	**End Date:**	03/31/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 3/31/2021 as grant ended early per PI move (Ed., 8/9/21)		
Key Personnel Changes/Previous PI:	Final report August 2021: The COVID-19 pandemic continued to hinder progress on this grant. Pandemic-imposed travel restrictions at the Noble Research Institute prevented us from traveling to the Microgravity Simulation Support Facility (MSSF) to complete several experiments proposed in specific aims 1 and 3. In addition to the problems brought about by COVID-19, the Noble Research Institute leadership discontinued its programs on plant science, plant breeding and laboratory-based science research. A fallout from this decision was the termination of all external grants that were not aligned with the new focus of the Institute on regenerative ranching. This NASA grant was one of several federal grants at the Noble Research Institute that were asked to be discontinued. Furthermore, the PI (Blancaflor) and Co-I (Nakashima) ended their employment at the Noble Research Institute in March and May, respectively, as a result of the decision to terminate basic plant science research. Therefore, on March 2021, a request was made to NASA to transfer remaining work on this grant to Dr. Simon Gilroy at the University of Wisconsin. Once the grant is transferred to		

	Wisconsin, a no cost extension (NCE) will be requested so that remaining tasks in aims 1 and 3 could be completed. PI change: Dr. Simon Gilroy will be the new PI at Wisconsin). Other key personnel change Dr. Sabrina Chin (former postdoctoral fellow at Noble who will continue as a postdoc in the Gilroy lab at Wisconsin).
COI Name (Institution):	Nakashima, Jin Ph.D. (Noble Research Institute)
Grant/Contract No.:	80NSSC18K1462
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Plants will be a major component of advanced life support systems to enable NASA to realize its vision of long duration space exploration that will take humans to Mars and beyond. However, plants can only be effectively used for such systems if we have a deeper mechanistic understanding about how their development is regulated by the spaceflight environment. Recent "omics" studies have shown that plants reprogram their genetic circuitry when grown in the near weightless conditions of space. Genes involved in cell wall remodeling and oxidative stress are major gene categories that are differentially regulated by microgravity. Further, it has been shown that microgravity in space and simulated microgravity using clinostats or random positioning machines elicit roughly similar root skewing responses in the model plant Arabidopsis thaliana. Root skewing was more pronounced in mutants to a root-expressed vegetative actin isoform. Consistent with the enhanced root skewing of actin mutants in microgravity is the observation that other plant species such as maize (Zea mays L.) have stronger root directional growth responses on a clinostat when chemicals such as latrunculin B disrupt their actin cytoskeleton. Based on these observations, we will test the hypothesis that the actin cytoskeleton is a key player in dictating root growth directionality in microgravity and simulated microgravity by regulating the process of autotropic organ straightening. Moreover, we hypothesize that a set of cell wall cross-linking (i.e., class III peroxidases) and putative actin-regulatory proteins (i.e., HLB1 and MIN7/BEN1) mediate root hair elongation and directional root growth, respectively, under microgravity and simulated microgravity. Our hypotheses will be addressed through three specific aims including the analysis of cell wall composition/distribution in wild-type Arabidopsis roots, and detailed characterization of selected Arabidopsis cell wall and actin mutants subjected to microgravity simulation on a slow rotating clinostat and random position machine (aim 1 and 3). We will also use microgravity simulators to address the hypothesis that the plant hormone brassinolide regulates directional root growth in maize through its modulation of actin organization and dynamics (aim 2). Our research plan builds on successes of previous NASA-funded ground and spaceflight results, and outlines a comprehensive set of experiments including cell wall immunocytochemistry, computer-based root growth analyses, actin quantification, and genetics. The proposed work aligns with Plant Biology (PB) Element of the Space Biology Science Plan. Results obtained from plant experiments on microgravity simulators will address guiding questions linked to the decadal survey recommendations including the impact of gravity on plant growth, development and metabolism (PB-1), and molecular mechanisms underlying how plants sense and react to gravity (PB-5). We expect that the proposed research will provide us with basic knowledge that will guide the development of plant cultivars better adapted to the microgravity environment of spaceflight, an outcome that will prove beneficial for the design of plant habitats on future space colonies and for addressing great agricultural challenges on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposal aims to gain a better mechanistic understanding of how plant roots function at the cellular and molecular levels. A primary focus of the proposed work is to ask how the microgravity environment modulates the directionality of root growth. Because roots are essential for the uptake of water and nutrients, knowledge gained from successful implementation of the proposed work will be crucial in guiding pre-breeding efforts aimed toward the development of crops with improved productivity through better and more efficient root systems. Thus, agriculture on Earth is one beneficial outcome of this project. Innovative technologies developed through this grant include lines of new model plants such as barrel medic (Medicago truncatula) and purple false brome (Brachypodium distachyon) expressing genetically-encoded sensors to the cytoskeleton and cytoplasmic calcium. We expect that these new plant lines will significantly advance basic plant biology research on Earth.
	Although progress was made on all 3 specific aims of this proposal, it was limited because of the continuing COVID-19 pandemic and closure of the PI's laboratory at the Noble Research Institute. Below we summarize progress for each aim. For aim 1, most of our progress was made on processing Arabidopsis thaliana seedlings for immunocytochemistry. Because we were not able to travel to the Microgravity Simulation Support Facility (MSSF) at the Kennedy Space Center in 2020, we focused our work activities in this aim using a custom-made 2D clinostat. Four-day old seedlings of Arabidopsis thaliana grown in the light and dark were mounted on the clinostat, and rotated in an axial direction for 2-weeks. Corresponding 1-g controls were included in the experiment. A 2-week clinorotation regimen was selected to approximately mirror the period of time seedlings in our previous Biological Research In Canisters (BRIC) BRIC-16 and Advanced Plant Experiments (APEX) APEX-03-1 experiments were exposed to microgravity on the space shuttle and International Space Station (ISS), respectively. Each treatment condition had four 10 cm x 10 cm square Petri dishes. After 2 weeks of growth, at least 5 seedlings for each condition (i.e., clinorotated and 1-g controls) were preserved in an aldehyde solution and processed for flat embedded in LR White resin. The remaining seedlings were kept in Phosphate Buffered Saline (PBS). Because of the closure of the PI’s lab at Noble, embedded and PBS-stored samples were shipped to the laboratory of Dr. Simon Gilroy (University of Wisconsin). Future studies in the Gilroy lab upon transfer of the grant will involve sectioning the embedded seedlings and subsequent immunocytochemistry of selected non-cellulosic cell wall components. For specific aim 2, we have successfully generated lines of Brachypodium distachyon (a model grass) expressing green fluorescent protein (GFP)-tagged microtubule and actin reporters. We expect that these live cell cytoskeletal reporters in a model grass species will advanced studies of grass cell biology. Through past and current NASA support, we have now generated plant lines in Arabidopsis thaliana, Medicago truncatula, and Brachypodium distachyon expressing live cell microtubule and actin reporters. All these lines have been sent to the Gilroy lab in Wisconsin for use in future studies of microgravity- and clinorotated-induced changes in the cytoskeleton.

Task Progress:	Most of our progress for aim 3 has been on characterizing actin cytoskeletal-related mutants that we isolated from previous NASA-funded research (APEX 03-1). Two peer-reviewed articles relevant to this aim were published during this reporting period. One noteworthy accomplishment during this reporting period was completing detailed cell biological studies on the recessive Arabidopsis thaliana hypersensitive to LatB2 (hlb2) mutant. Through map-based cloning, we found that the mutation in hlb2 had a 10 base pair deletion in exon 14 of the AT1G03060 gene. AT1G03060 encodes a protein called SPIRRIG (SPI), a 3,571 amino acid long protein with N-terminally located armadillo (ARM) and concanavalin A (ConA)-like lectin domains and C-terminally-located pleckstrin homology (PH), beige and Chediak Higashi (BEACH) and WD40 repeat domains. BEACH domain-containing proteins are highly conserved in eukaryotes and are known to function in membrane dynamics, vesicle transport, apoptosis and receptor signaling. In a recent paper published in The Plant Cell, we studied root hair development in the hlb2/spi mutant. Root hairs are tip growing cells (i.e., cell expansion is confined to the apical-most domain of the cell) that extend the effective radius of a root to enable plants to optimize water and nutrient absorption. In previous spaceflight studies, we found that root hair development is impaired in microgravity. Root hair development is a complex process that involves the actin cytoskeleton and a multitude of proteins that specify the position of root hair emergence, and drive tip-focused growth. To gain an in-depth understanding of the role of actin during root hair development, it is important to be able to visualize the dynamics of proteins that may directly or indirectly modulate actin organization during this process. Visualizing protein dynamics in living cells is typically accomplished by tagging with a fluorescent protein such as GFP. However, some proteins that modulate root hair development are large and of low abundance, and therefore more challenging to analyze when tagged with fluorescent proteins. SPI is one such protein as it is 3,571 amino acids long. In the absence of SPI, root hair tip growth and the position of root hair emergence is stunted suggesting that this protein is a major modulator of root hair development. In the aforementioned Plant Cell publication, we generated a fully functional fluorescent protein for SPI using a technique called recombineering in the genome of Arabidopsis thaliana. Expressing the SPI protein fusion in the hlb2/spi mutant drove normal root hair development and localized to the tip-most domain of rapidly elongating root hairs. The root hair tip localization of SPI was correlated with tip-focused filamentous-actin (F-actin) meshwork . Our work also revealed a link between SPI and selected components of the actin nucleating promoting complex WAVE/SCAR (W/SCR). For example, a fully functional BRICK1 (BRK1) fluorescent protein fusion (BRK1-mRuby3), a subunit of W/SCR, marked the position of root hair emergence. Live cell microscopy of BRK1-YFP and SPI-YPet revealed contrasting spatial and temporal dynamics, with the former intensifying and the latter dissipating as root hairs elongated. To observe BRK1 and SPI simultaneously within the same root hair, we generated plants expressing both SPI-YPet and BRK1-mRuby3. BRK1-mRuby3 labeled the site in which a new root hair emerged and dissipated as the root hair bulge expanded. Within the same root hair cell, SPI-YPet fluorescence at the tip intensified as BRK1-mRuby3 dissipated. The weakening of BRK1-mRuby3 fluorescence coinciding with SPI-YPet accumulation provides indirect evidence that SPI might play a role in mediating BRK1 stability or localized clustering at the plasma membrane at the position of root hair emergence. Although it is unknown why BRK1-YFP signal persists in spi, we speculate that SPI might modulate BRK1 via protein degradation pathways .Our work on HLB2/SPI provides new insights that contribute to our understanding of actin-mediated root hair development, and point to potential functional relationships among HLB2/SPI and actin nucleation proteins during root hair development. For the future, we plan to evaluate genetic interactions between SPI and components of actin-related protein 2/3 (ARP2/3)-W/SCR complex. The live cell actin reporters we generated through NASA funding has also resulted in productive research collaborations with external Institutions. One noteworthy collaboration that made extensive use of our live cell actin reporters was with the group of Dr. Yangdou Wei at the University of Saskachewan, Canada. Using Arabidopsis thaliana lines expressing the GFP-ABD2-GFP live cell actin reporter, new details on how plants defend themselves against the powdery mildew fungal pathogen was discovered. It was shown that one of the earliest responses of plants to powdery mildew was the formation of actin patches beneath the fungal invasion sites. The assembly of this actin patch is facilitated by the ARP2/3-W/SCR complex and another actin nucleating protein called formin. NOTE: The PI (Blancaflor) and Co-I (Nakashima) ended their employment at the Noble Research Institute in March and May, respectively, as a result of the Noble decision to terminate basic plant science research. Therefore, on March 2021, a request was made to NASA to transfer remaining work on this grant to Dr. Simon Gilroy at the University of Wisconsin and Dr. Gilroy will be new PI for this work.

Bibliography Type:	Description: (Last Updated: 01/13/2026)
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